

E-Commerce Sales Analysis Report

A Data Analytics Project Using Python

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PROJECT INTRODUCTION

This project analyzes an e-commerce sales dataset using Python. The goal is to understand sales performance, customer behavior, product demand, payment patterns, referral source performance, order status distribution, coupon usage, and revenue trends over time.

The analysis was performed using Python libraries such as pandas, matplotlib, and seaborn. The dataset was cleaned, summarized, grouped, and visualized to produce a simple business intelligence dashboard.

PROJECT SUMMARY

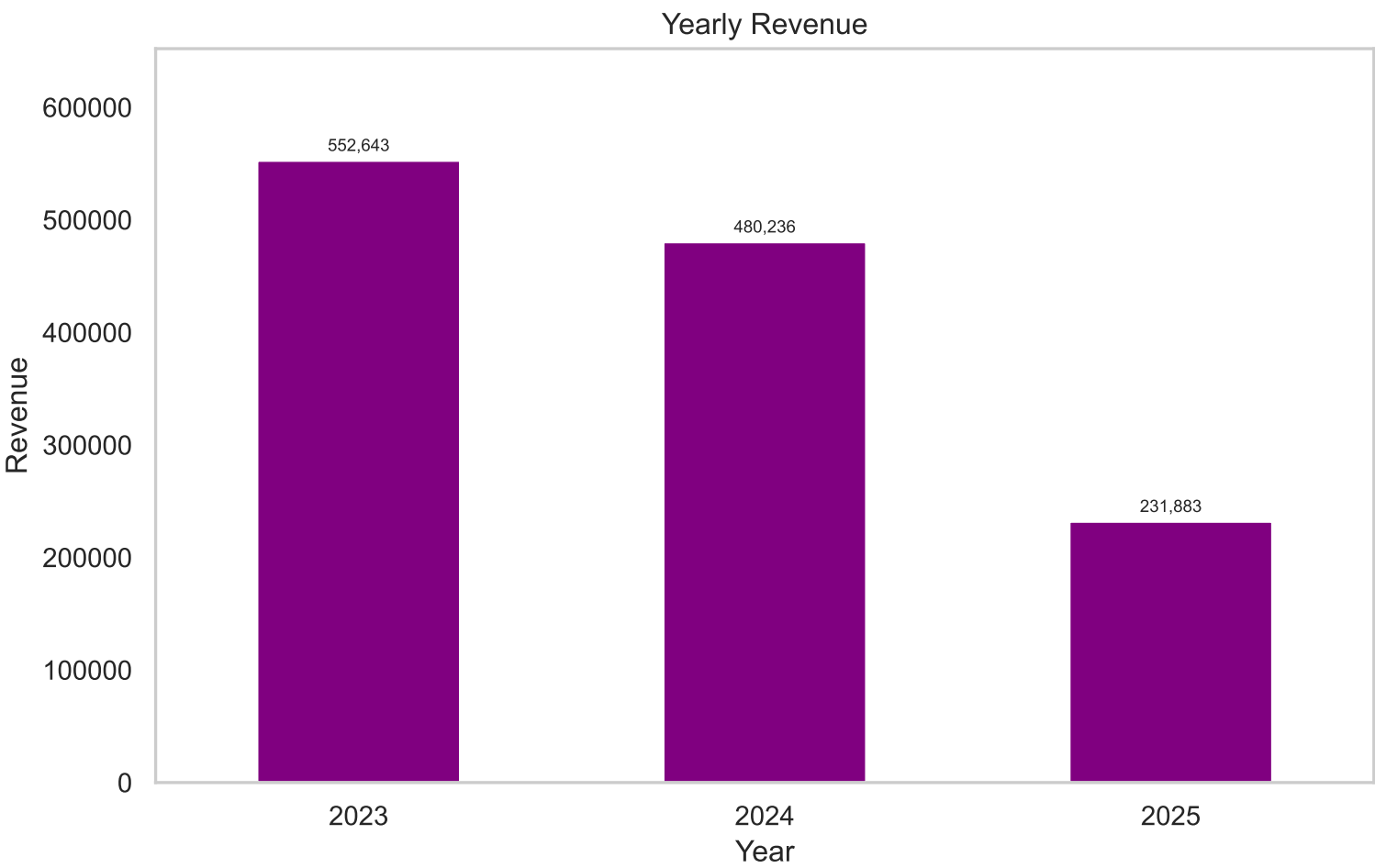
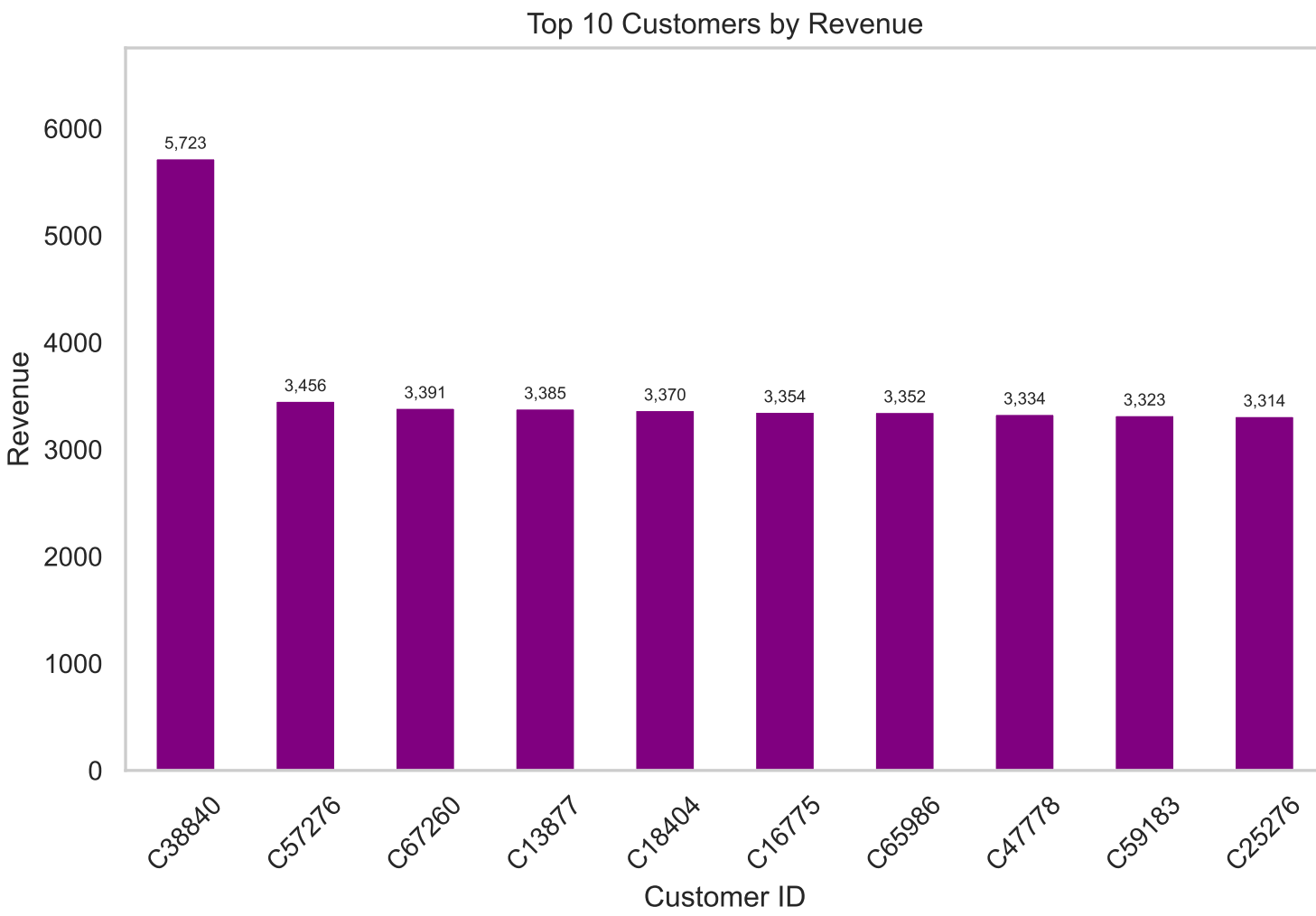
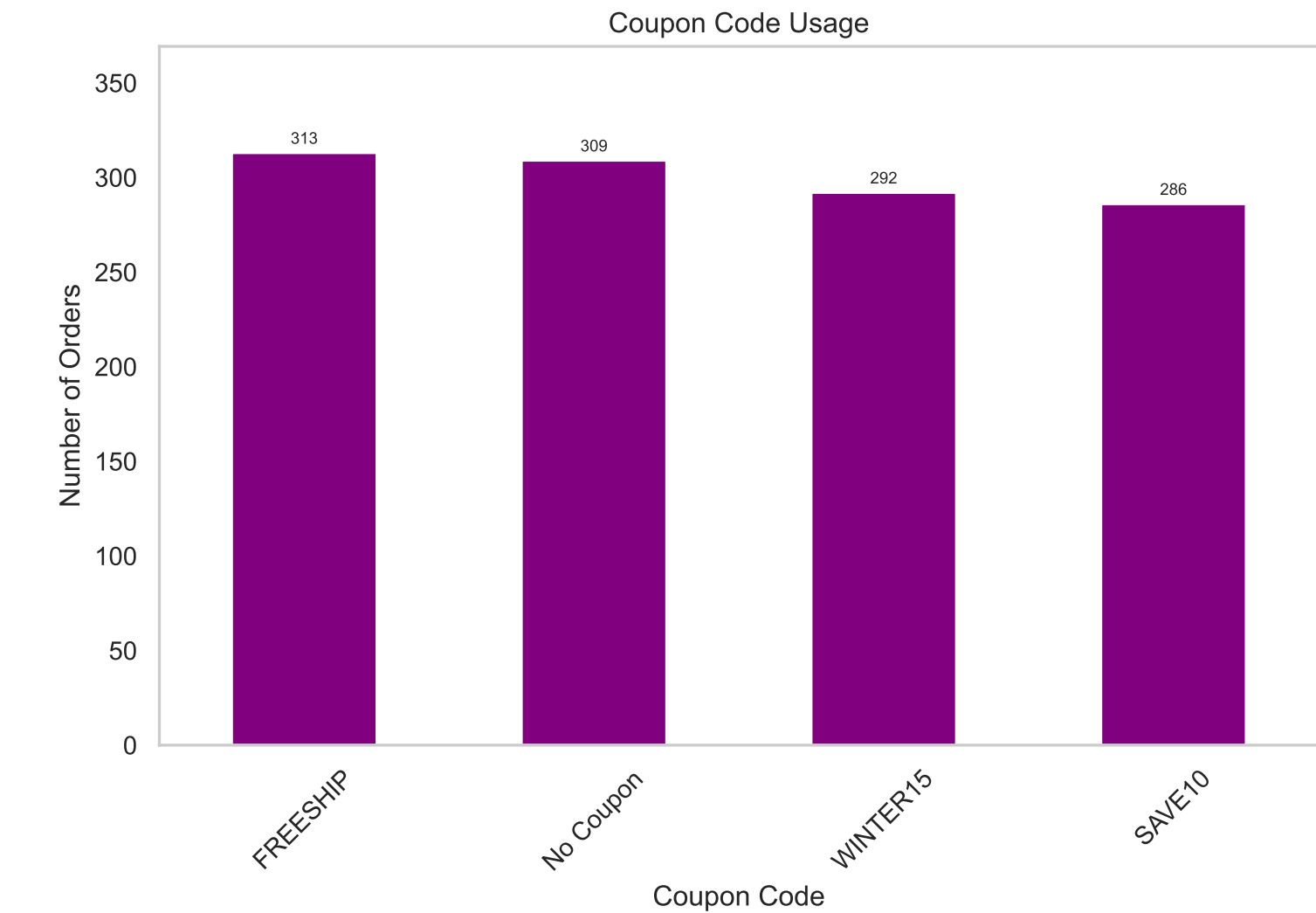
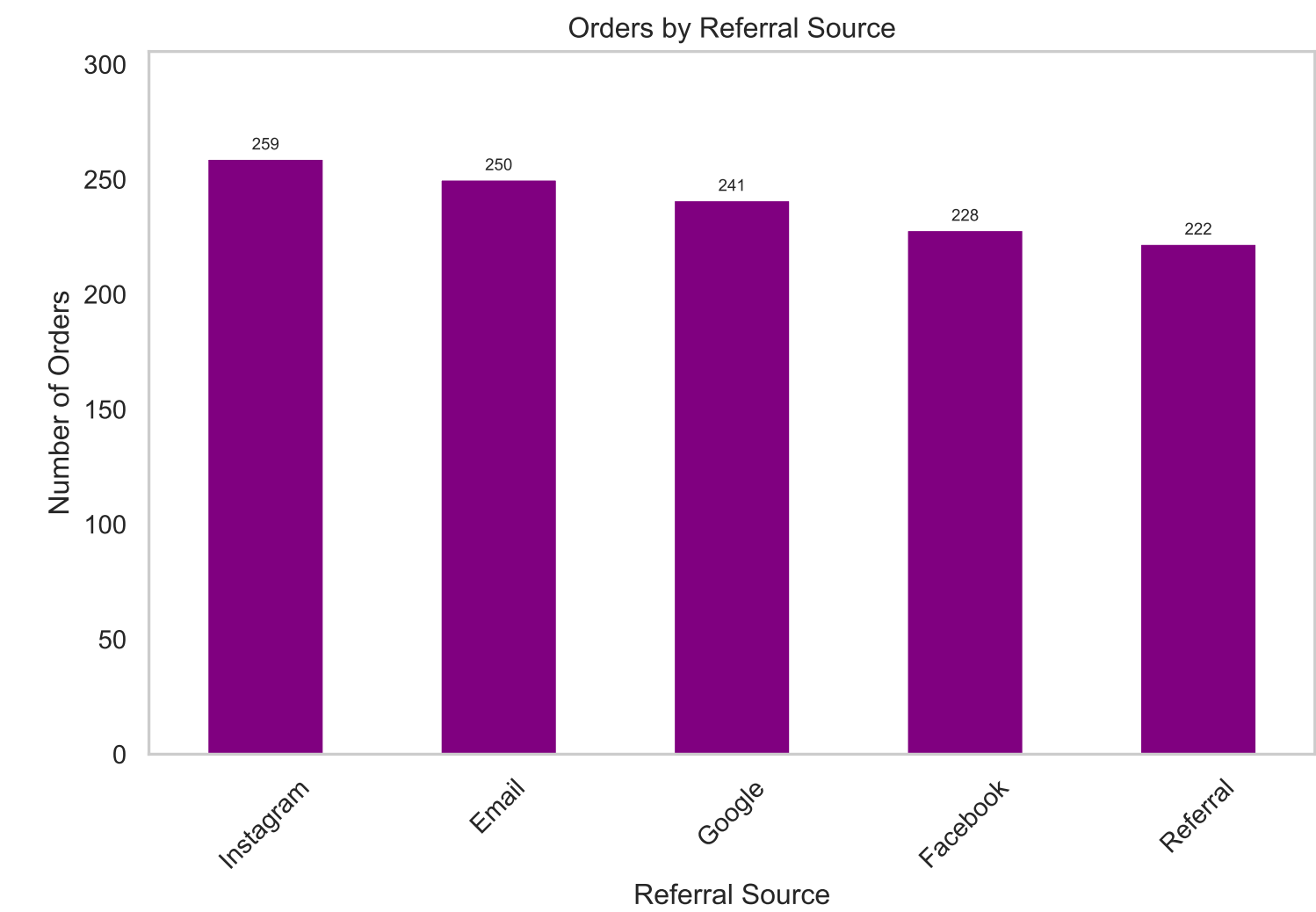
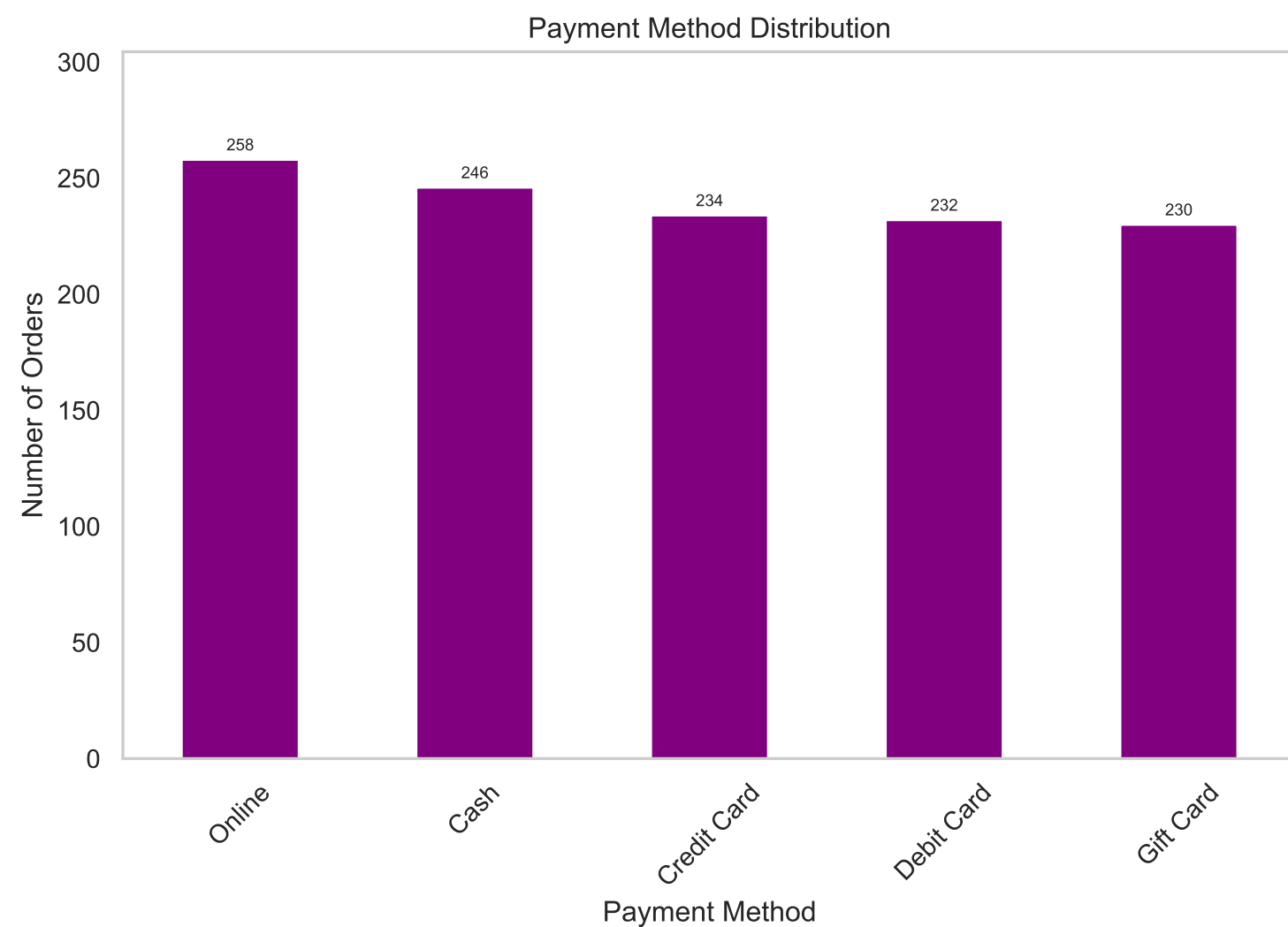
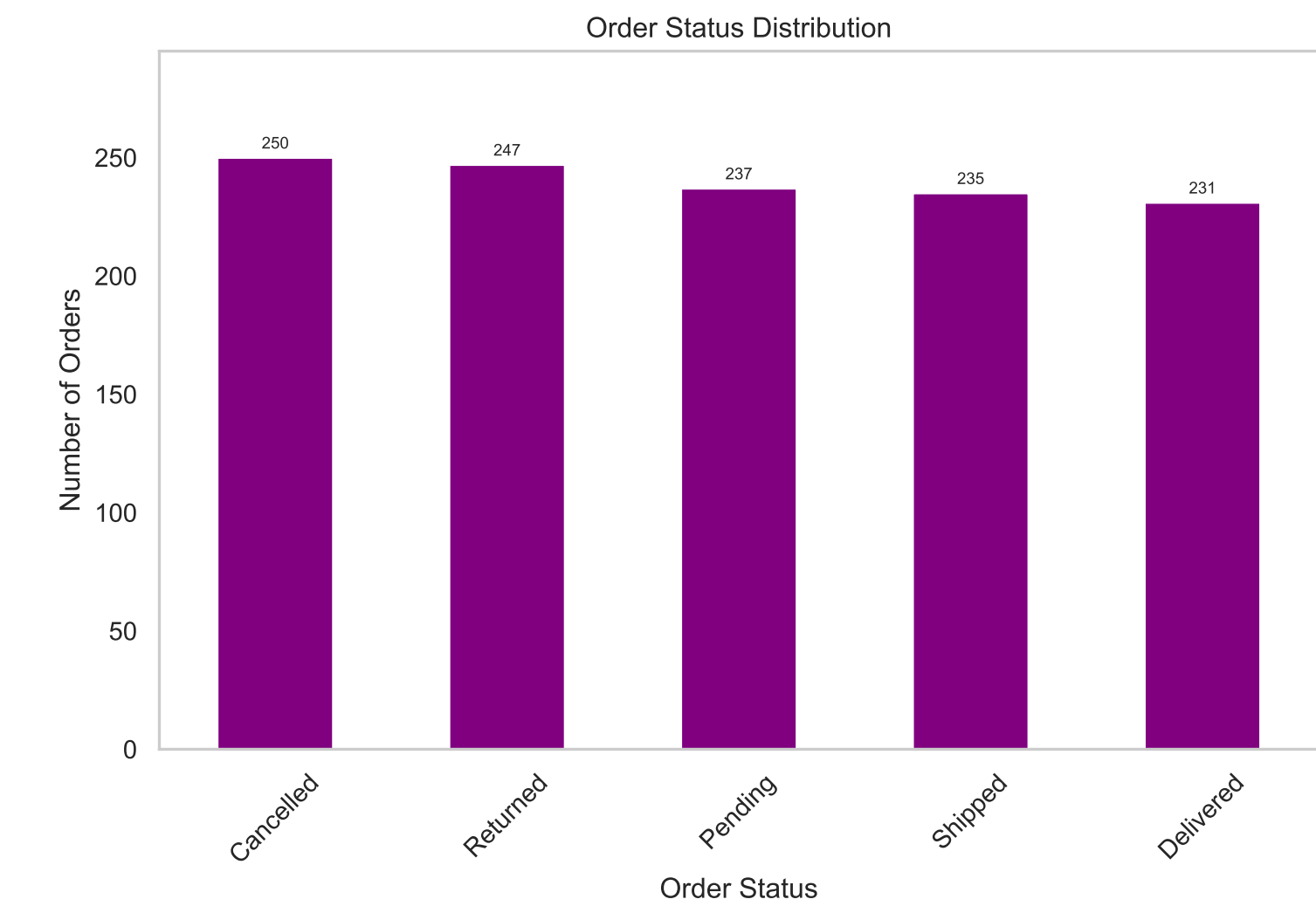
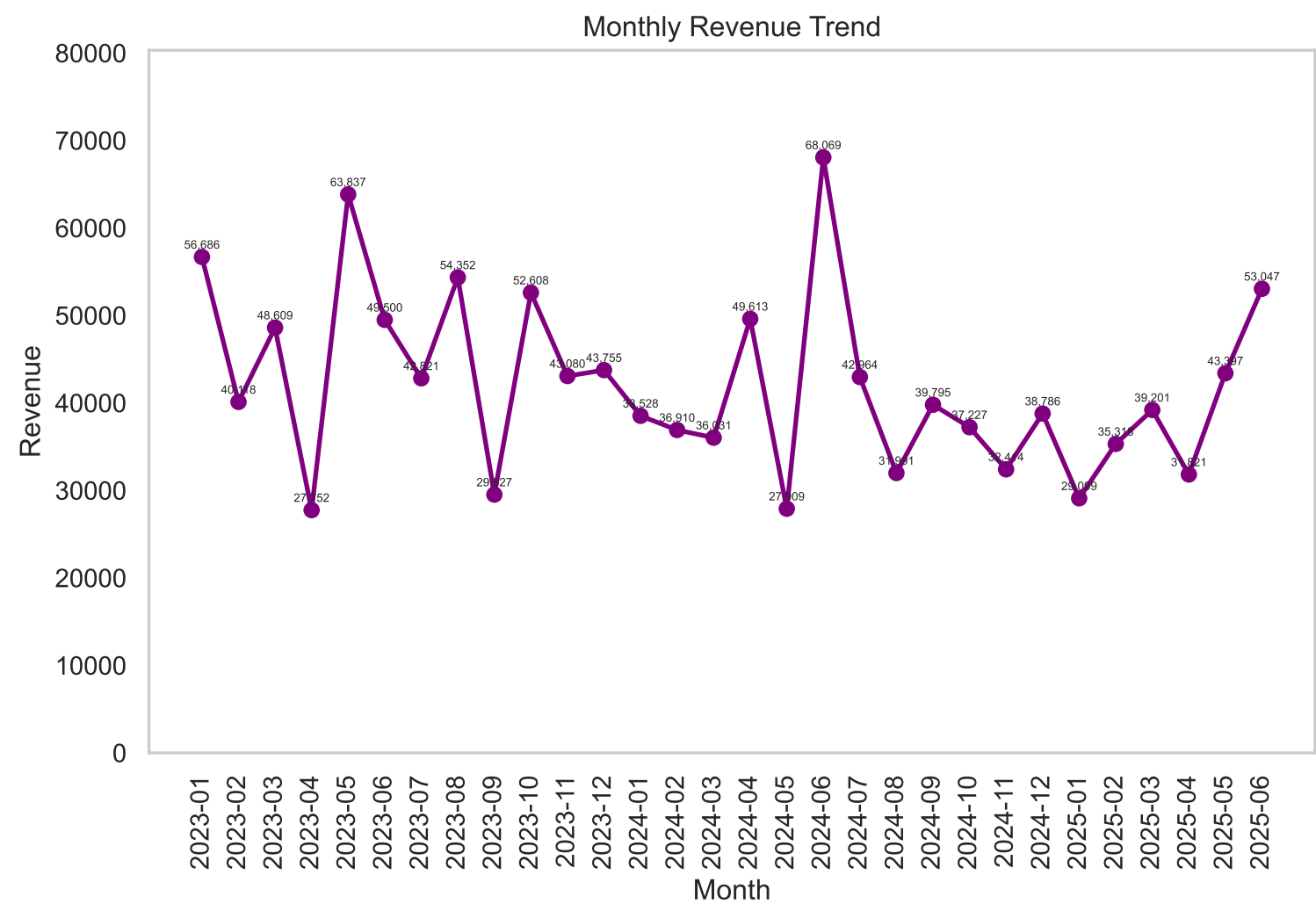
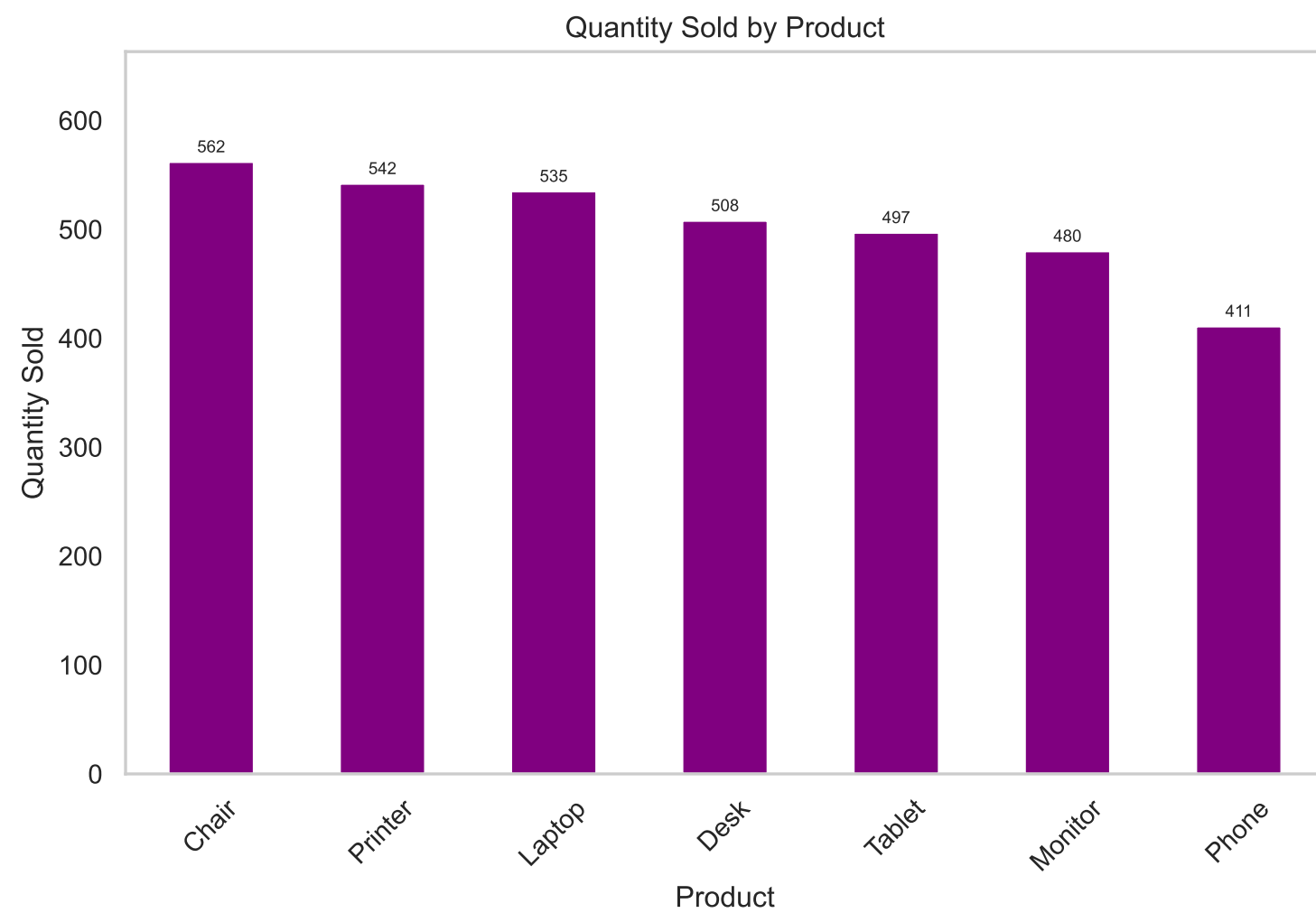
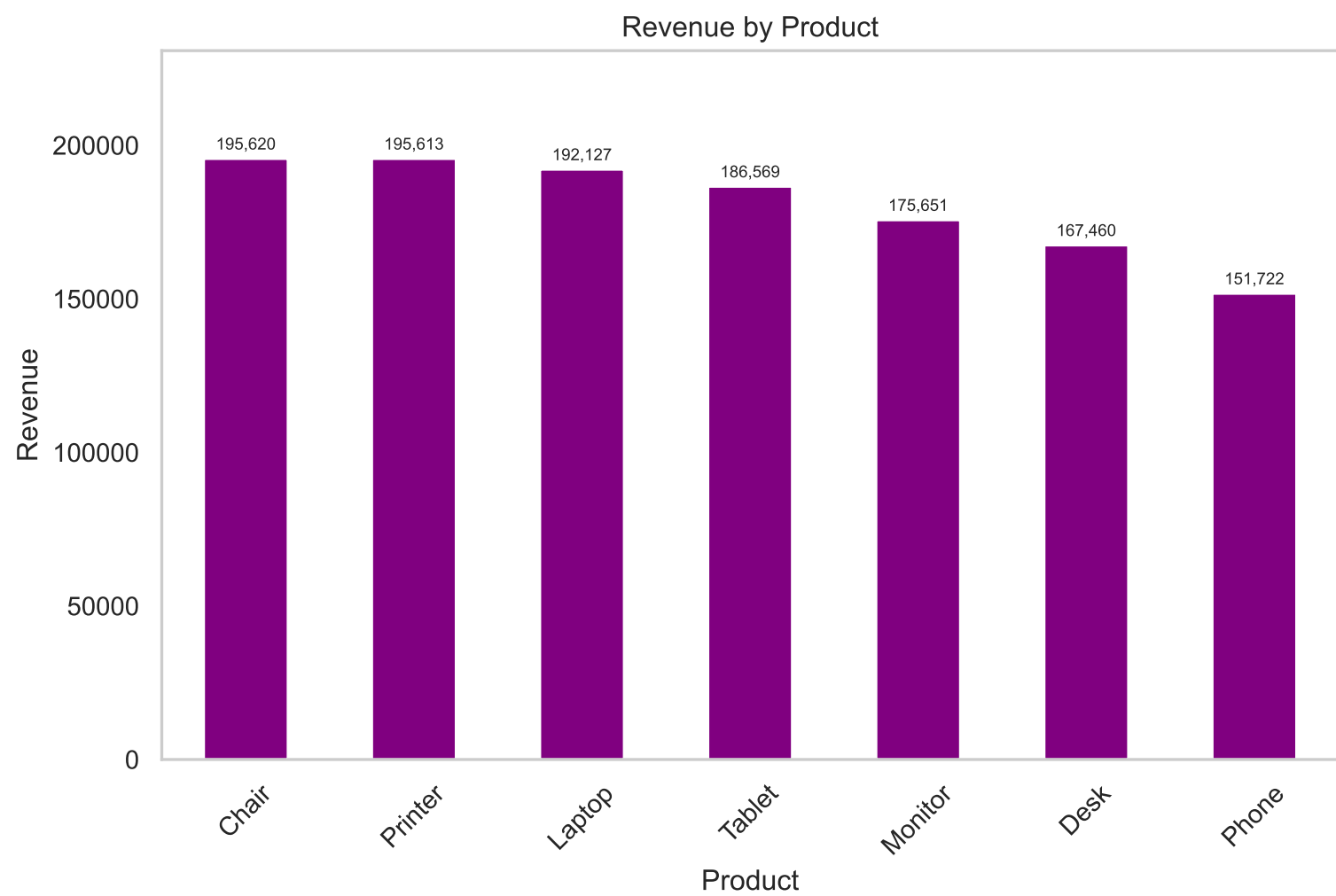
The dashboard contains nine visuals covering the most important areas of the business:

1. Revenue by Product
2. Quantity Sold by Product
3. Monthly Revenue Trend
4. Order Status Distribution
5. Payment Method Distribution
6. Orders by Referral Source
7. Coupon Code Usage
8. Top 10 Customers by Revenue
9. Yearly Revenue

Purple was used as the consistent chart color to maintain a clean visual style. Gridlines were removed to make the charts easier to read, and values were added directly to the bars and line chart points for clearer interpretation.

The dashboard helps identify the strongest revenue drivers, best-selling products, customer contribution, sales trends, and business channels that perform well.

E-Commerce Sales Analysis Dashboard



Product Performance Summary

Product	Total_Revenue	Total_Orders	Quantity_Sold	Average_Order_Value
Chair	195620.11	178	562	1098.99
Printer	195612.61	181	542	1080.73
Laptop	192126.56	173	535	1110.56
Tablet	186568.95	179	497	1042.28
Monitor	175651.41	163	480	1077.62
Desk	167459.93	170	508	985.06
Phone	151722.39	156	411	972.58

Business Summary Tables

Payment Method Summary

PaymentMethod	Total_Revenue	Total_Orders	Average_Order_Value
Credit Card	263847.63	234	1127.55
Online	262442.94	258	1017.22
Cash	259786.29	246	1056.04
Gift Card	246323.92	230	1070.97
Debit Card	232361.18	232	1001.56

Referral Source Summary

ReferralSource	Total_Revenue	Total_Orders	Average_Order_Value
Instagram	275285.45	259	1062.88
Email	261808.55	250	1047.23
Google	250441.48	241	1039.18
Facebook	250410.9	228	1098.29
Referral	226815.58	222	1021.69

Order Status Summary

OrderStatus	Total_Revenue	Total_Orders	Average_Order_Value
Cancelled	276396.21	250	1105.58
Pending	256328.15	237	1081.55
Shipped	246159.58	235	1047.49
Returned	243277.7	247	984.93
Delivered	242600.32	231	1050.22